

### 1 . Introduction to Pandas

- 2 . Library for data manipulation & analysis.
- 3 . Main data structures: Series ( 1 D), DataFrame ( 2 D).

```
import pandas as pd
s = pd.Series([10, 20, 30], index=['a', 'b', 'c'])
df = pd.DataFrame({'Name': ['Alice', 'Bob'], 'Age': [25, 30]})
```

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### 1 . Creating DataFrames

- 2 . From dict: `pd.DataFrame({'A': [1, 2], 'B': [3, 4]})`
- 3 . From list of lists: `pd.DataFrame([[1, 2], [3, 4]], columns=['A', 'B'])`
- 4 . From NumPy: `pd.DataFrame(np.arange(6).reshape(2, 3), columns=['A', 'B', 'C'])`
- 5 . From CSV/Excel: `pd.read_csv('file.csv')`, `pd.read_excel('file.xlsx')`

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### 1 . Data Inspection

- 2 . `head()`, `tail()`, `shape`, `info()`, `describe()`

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### 1 . Selecting Data

- 2 . Columns: `df['Name']`, `df[['Name', 'Age']]`
- 3 . Rows: `df.iloc[0]`, `df.iloc[0:2]`, `df.loc[0, 'Name']`

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### 1 . Filtering & Conditional Selection

```
df[df['Age'] > 30]
df[(df['Age'] > 25) & (df['City'] == 'Dhaka')]
```

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### 1 . Adding/Modifying Columns

```
df['Salary'] = [50000, 60000]
df['Age'] = df['Age'] + 1
```

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### 1 . Handling Missing Data

2. `isnull()`, `dropna()`, `fillna()`
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### 1 . Sorting & Ranking

2. `sort_values()`, `rank()`
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### 1 . Grouping & Aggregation

```
grouped = df.groupby('City')
grouped['Salary'].mean()
grouped['Age'].agg(['min', 'max', 'mean'])
```

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### 1 . Merging & Joining

```
df1 = pd.DataFrame({'ID':[1,2], 'Name':['Alice', 'Bob']})
df2 = pd.DataFrame({'ID':[1,2], 'Salary':[50000, 60000]})
merged = pd.merge(df1, df2, on='ID')
```

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### 1 . Pivot Tables

```
df.pivot_table(index='City', values='Salary', aggfunc='mean')
```

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### 1 . Input/Output

2. `to_csv()`, `read_csv()`, `to_excel()`, `read_excel()`
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### 1 . Practice Exercises

- 2 . Create DataFrame of students with Name, Age, Grade, City.
  - 3 . Filter students older than 20.
  - 4 . Add column `Passed` (Grade > 50 → True).
  - 5 . Sort by `Grade` descending.
  - 6 . Group by `City` and find average `Age`.
  - 7 . Merge two DataFrames on a common column.
  - 8 . Handle missing values.
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### 1 . Interview Questions

- 2 . Series vs DataFrame?

- 3 . Handling missing data?
- 4 . `loc` vs `iloc` ?
- 5 . How `groupby` works?
- 6 . Merge/join DataFrames?
- 7 . Pivot vs Pivot\_table?
- 8 . Apply function to column/row? ( `apply` )